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Research on femtosecond laser welding process of silver nanowire transparent electrodes

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ABSTRACT

With the continuous development of flexible transparent electrodes, silver nanowires have been widely used due to their excellent optoelectronic properties and flexibility. However, their progress has been hindered by high contact resistance. In this study, femtosecond laser post-processing was applied to the silver nanowire transparent electrodes, and the effect of laser power density on the welding of silver nanowires was investigated. The results showed that femtosecond laser could achieve welding between silver nanowires, but excessively high laser energy density could cause the melting of the nanowires. When the femtosecond laser energy density reached the critical energy density of approximately 2546 $J \cdot m^{-2}$, at which nanowires underwent spheroidization, the average sheet resistance of the silver nanowire electrodes was minimized to 18.89 $\Omega \cdot sq^{-1}$, and the figure of merit (*FoM*) of the electrodes reached its maximum at 126.99, which was close to twice the initial value. Moreover, the non-uniformity factor (*NUF*) was also small, at 1.6%, indicating good conductivity and uniformity. Additionally, the influence of femtosecond laser irradiation on the transmittance of the silver nanowire transparent electrodes was insignificant, and the number of effective pulses had no obvious effect on the welding efficiency of the silver nanowires.

1. Introduction

With the advancement of technology, electronic devices are expected to become more flexible in the future. This includes the development of flexible wearable electronic devices, electronic skin, touch screens, and flexible solar cells, among others. These new types of electronic devices possess excellent folding and stretching capabilities. Flexible transparent electrodes, serving as a vital component in these devices, have garnered substantial attention both domestically and internationally [1].

Currently, the most widely used transparent electrode material is indium tin oxide (ITO) [2], but it has poor mechanical flexibility [3] and high fabrication costs. AgNW(silver nanowire)

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transparent electrodes present a promising alternative to ITO. The AgNW transparent electrode is esteemed for its exceptional conductivity and ductility. As a representative one-dimensional nanostructured material, AgNWs display minimal internal defects in comparison to other materials [4]. Moreover, AgNWs exhibit a relatively low melting point, allowing for processing at lower temperatures [5]. This property enables better compatibility with the ambient or lower temperature preparation environment required for polymer substrates.

However, the interconnection method between AgNWs is based on a contact-based junction, which is not reliable and can lead to significant contact resistance. There is considerable room for improvement in the conductivity of AgNW electrodes [6–8]. This issue impedes the large-scale application of AgNWs as flexible transparent electrodes. In order to enhance the conductivity of AgNW electrodes, it is typically necessary to perform post-treatments that strengthen the interconnection between the AgNWs. The post-processing methods mainly include heating, pressure application, electron beam welding, chemical treatment, and irradiation.

The most commonly used method is thermal annealing, which significantly reduces the sheet resistance of AgNW electrodes after heat treatment. However, excessively high heating temperatures and long heating times can lead to extensive melting and rupture of AgNWs. Furthermore, there is a risk of irreversible damage to flexible substrates, which can impact the stability of the conductive films [9]. Pressure bonding flattens the overlapping regions of AgNWs by uniformly applying pressure, thereby increasing the contact area between the nanowires. This reduces both the contact resistance and the roughness of the AgNW electrodes. However, pressure bonding struggles to achieve atomic-level contact between the crystal lattice planes and can unpredictably damage the substrate [10,11]. Joule heating method directly passes current through the nanowires, generating Joule heating at the contact points, which melts the contact points and achieves connection of the nanowire conductive film [12]. However, this method requires a uniform distribution of sheet resistance in the nanowire film. Otherwise, unstable currents caused by varying local resistances can lead to localized high temperatures and damage to the flexible substrate. High-energy electron beam welding utilizes the thermal effect of electron inelastic scattering to achieve the welding of metal nanowires [13]. However, it requires a vacuum environment and precise alignment of nanowires for successful welding. Which significantly increases the difficulty and cost of the processing process. The chemical treatment involves using the prepared AgNW electrode as the cathode, which is submerged in a solution containing metal salts and buffering agents, while a metal plate is used as the anode. Under the influence of electric current, both the surfaces of the nanowires and their junctions undergo reduction of metal ions, thereby further connecting the AgNWs. However, this method may result in the widening and thickening of nanowires, resulting in increased light absorption and scattering, and hence significant transparency losses [14].

Irradiation method utilizes the local electric field enhancement generated by the localized electron resonance excitation of plasmon achieved in the gap between AgNWs by incident light electromagnetic field. It enables reliable connection of nanomaterials without causing damage to heat-sensitive materials such as flexible substrates, greatly improving the optoelectrical properties and stability of conductive films. Chung et al. achieved reliable connection of AgNWs using ultraviolet and white light, ultimately obtaining AgNW electrodes with good mechanical properties, sheet resistance of 77.93 $\Omega\cdot sq^{-1}$, and transmittance of 98.76% [15]. Jiu et al. used microsecond pulse laser and obtained AgNW electrodes with sheet resistance of 19 $\Omega\cdot sq^{-1}$ and transmittance of 83% [16]. Spechler et al. used nanosecond laser irradiation on AgNWs, resulting in a 75% reduction in sheet resistance to 3–6 $\Omega\cdot sq^{-1}$ and transmittance ranging from 55% to 70% [17]. However, long-pulse lasers such as nanosecond and microsecond lasers generate significant heat effects during irradiation, which can lead to surface roughening or agglomeration of AgNWs and substrate damage and deformation. Therefore, it is necessary to find a laser light source with lower thermal effects.

Femtosecond laser is a unique tool for material processing, operating in pulse mode with a small focusing diameter and an extremely short duration in the femtosecond range (10^{-15} s). It delivers instantaneous high power and induces highly non-equilibrium states and non-thermal phase transitions when applied to materials [18,19]. Due to its short duration, the thermal effects generated by femtosecond laser processing can be frozen near the initial energy deposition location, enabling localized heating [20]. Studies have shown that nanoparticles can melt at temperatures below their conventional melting points, which provides the conditions for nanoscale connections of AgNWs.

Ha et al. used a femtosecond laser with a wavelength of 800 nm to irradiate AgNWs on a PET substrate [21]. No AgNW rupture or thermal accumulation occurred, and the AgNWs were only connected at the overlapping regions, preserving their original crystalline structure. Eventually, they obtained AgNW electrodes with a sheet resistance of 25 $\Omega\cdot sq^{-1}$ and a transmittance exceeding 94% at 550 nm, demonstrating the feasibility of connecting AgNWs using femtosecond laser.

Hu et al. employed femtosecond laser with an energy density of 0.8–1.2 $\mu J/cm^2$ to weld AgNW flexible transparent conductive films (FTCFs), with a processing speed set at 1 mm/s^2 [22]. The results revealed that the non-uniformity coefficient of sheet resistance was 4.6% at an average sheet resistance of 16.1 $\Omega\cdot sq^{-1}$ and a transmittance of 91%. Additionally, the laser-welded AgNW FTCFs exhibited excellent performance during 10,000 cycles of mechanical bending, effectively enhancing the conductivity, uniformity, and reliability of AgNWs.

Liang et al. utilized a line-shaped femtosecond laser beam with a length of 773 μm to irradiate AgNWs [23]. The laser energy density was set between 0.17 and 0.60 J/cm^2, and by adjusting the process parameters, they found that the optimal energy density was 0.52 J/cm^2 while the scanning speed was 6.14 mm^2/s. Compared to traditional spherical lens focusing irradiation welding, the welding efficiency could be improved by several tens of times, and the AgNW transparent electrode exhibited good conductivity and uniformity. However, Liang did not conduct in-depth research on the mechanism of the impact of femtosecond laser energy density and scanning speed on the welding results.

The feasibility and reliability of using femtosecond laser to weld AgNWs have been confirmed by current research. However, the current focus was still on the sheet resistance, uniformity, and transmittance, with a lack of exploration on the process parameters of femtosecond laser, including energy density and laser scanning speed.

In this study, AgNW transparent electrodes were post-processed using the optimized process of femtosecond laser irradiation. The influence of laser energy density on the morphology, sheet resistance, transmittance, and quality factors of AgNW electrodes was investigated. Furthermore, the effect of the number of effective laser pulses on the welding performance of AgNWs was studied by varying the laser scanning speed. By optimizing the process parameters, on one hand, nano-welding of AgNWs was achieved without causing significant damage, thus enhancing the overall performance of the AgNW electrodes. On the other hand, a relatively faster laser scanning speed was chosen to ensure higher processing efficiency.

2. Materials and methods

2.1. Materials

The AgNWs (MG-NW-S40) used in the experiment were purchased from Shanghai Maoguo Nano Technology Co., LTD., and the main parameters are shown in Table 1. The substrate of the AgNW transparent electrode was 20 mm × 20 mm × 1 mm transparent quartz glass.

2.2. Preparation of AgNW transparent electrode

The first step was to clean the glass substrate using Airlaid-Paper, followed by immersing the glass in a solution of ethanol and acetone for 15 min of ultrasonic cleaning. Subsequently, the glass was placed on a heating plate at 150 °C for 15 min. After cooling, the glass was dried with nitrogen gas. To remove any remaining organic residues on the glass surface, the glass was then placed in a UV ozone cleaner for 15 min.

The second step involved preparing the AgNW solution. Based on our previous experimental results, the optimized concentration of the AgNW solution was 2 mg/ml, and the spin-coating speed was set at 2000 rpm. Under this process condition, the AgNWs exhibited a moderate density on the substrate, uniform distribution in all directions, and minimal aggregation between the nanowires. The tested performance was highly stable, meeting the requirements for subsequent experiments on femtosecond laser welding of AgNWs [24]. The original AgNW solution had a concentration of 5 mg/ml, and concentrations of 2 mg/ml, was achieved by adding absolute ethanol at volume ratios of 2:3. Next, the solution was stirred for 1 h using a magnetic stirrer to ensure a uniform dispersion of the AgNWs.

The final step was to deposit the AgNWs onto the glass substrate. Using a pipette gun, 100 μL of the AgNW solution was dropped onto the glass, and the solution was evenly spread across the substrate. Next, we used the spin coater to perform spin coating on the glass substrate with the solution. The AgNW transparent electrode was obtained after allowing the solvent to naturally evaporate.

2.3. Femtosecond laser processing system

The laser used in this experiment was a femtosecond laser. The parameters are shown in Table 2.

Fig. 1 illustrates the setup of the femtosecond laser processing system. The femtosecond laser is initially generated by the laser transmitter. It then passes through the attenuator and the shutter before reaching the reflector, which directs it towards the galvanometer system. Eventually, the laser passes through the lens set and converges onto the sample.

The galvanometer system is a crucial component of the femtosecond laser processing system. It operates by utilizing computer signals to control the X-axis and Y-axis galvanometers. Through the movement induced by swing motors, the galvanometers are capable of deflecting the laser beam in both the X-axis and Y-axis. This mechanism enables precise manipulation of the laser focus, facilitating targeted movement along the desired processing path on the sample.

2.4. Experimental scheme of femtosecond laser welding of AgNWs

In order to ensure that the entire sample is irradiated by the laser with a spot diameter of 20 μm, an S-shaped array was utilized as shown in Fig. 2. The laser processing path in this experiment had an S-shaped configuration with a side length of 20 mm and a row spacing of 10 μm. This arrangement effectively covered the entire sample area. After determining the laser processing path, adjust the parameters such as laser energy density, scanning speed, and other relevant factors in the integrated software. Then, begin the irradiation processing of the AgNW electrodes.

2.5. Characterization

A scanning electron microscope (SU8010, Hitachi, Tokyo, Japan) was used to test the morphology of the samples. An

Table 2 – Main parameters of femtosecond pulsed laser.

Parameter	Value
Wavelength	1035 nm
Spot diameter	20 μm
Average output power	≥40 W
Output power range	0–40 W
Repetition frequency	Single-shot-4 MHz

Table 1 – Main parameters of silver nanowires (AgNWs).

Specification	Diameter nm	Length μm	Aspect Ratio	Purity %	Appearance	Concentration	melting point
MG-NW-S40	40 nm	45 μm	1125	99.9%	Incanus	5 mg/ml	523 K-573 K

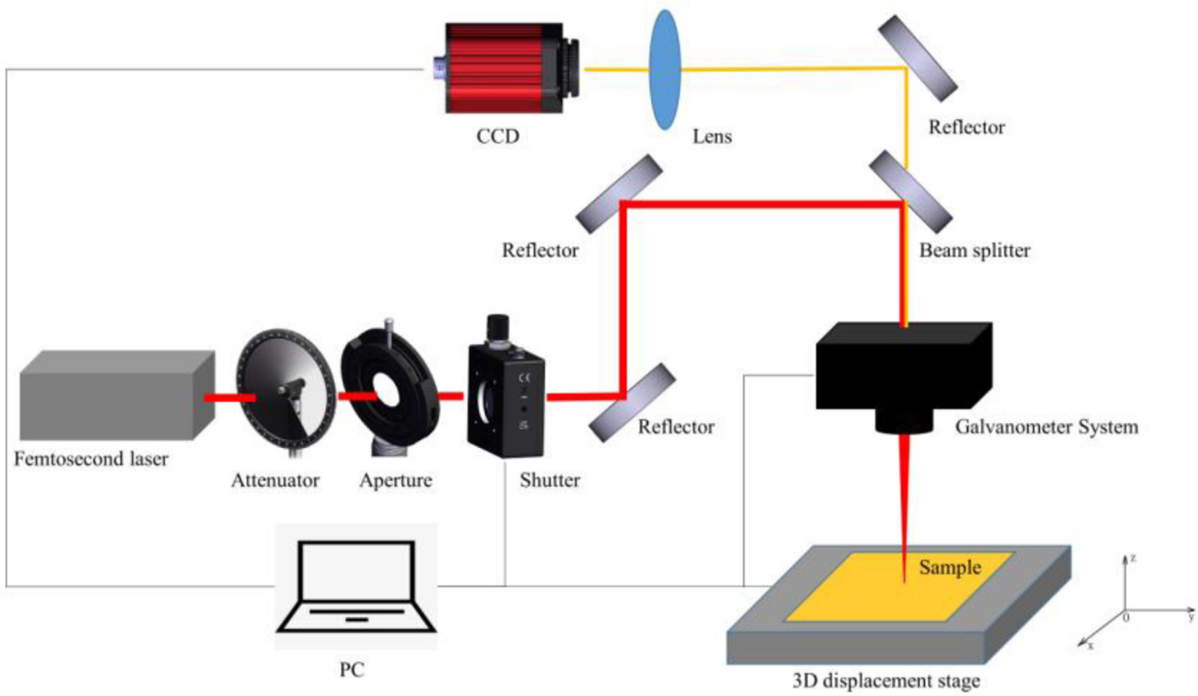

Fig. 1 – Schematic diagram of femtosecond laser system.

ultraviolet-visible-near-infrared spectrometer (UV3600 I Plus, Shimazu, Kyoto, Japan) was used to test the transmittance spectra of the samples. The sheet resistance (R_s) of the samples was tested using a four-point probe system (Model 280, Saratoga Technology, Saratoga, NY, USA).

3. Results and discussions

When using femtosecond laser for irradiation processing, the effective pulse number N must be considered. Which Represents the number of pulses received at the same location on the material during irradiation. Since the galvanometer scanning method is adopted in this experiment, the effective pulse number N can be calculated by the following equation.

$$N = Df/v \quad (1)$$

D, f and v respectively represent the laser spot diameter, laser frequency and galvanometer scanning speed. In this experiment, the value of D was 20 μm, while the value of f was 1 MHz. Therefore, the effective pulse number N was inversely proportional to the scanning speed v of the galvanometer. To ensure that all areas on the sample can be irradiated by the femtosecond laser, the effective pulse number N needs to be at least greater than 1.

If the effective pulse number N is too low, it may not be possible to achieve sufficient welding between AgNWs.

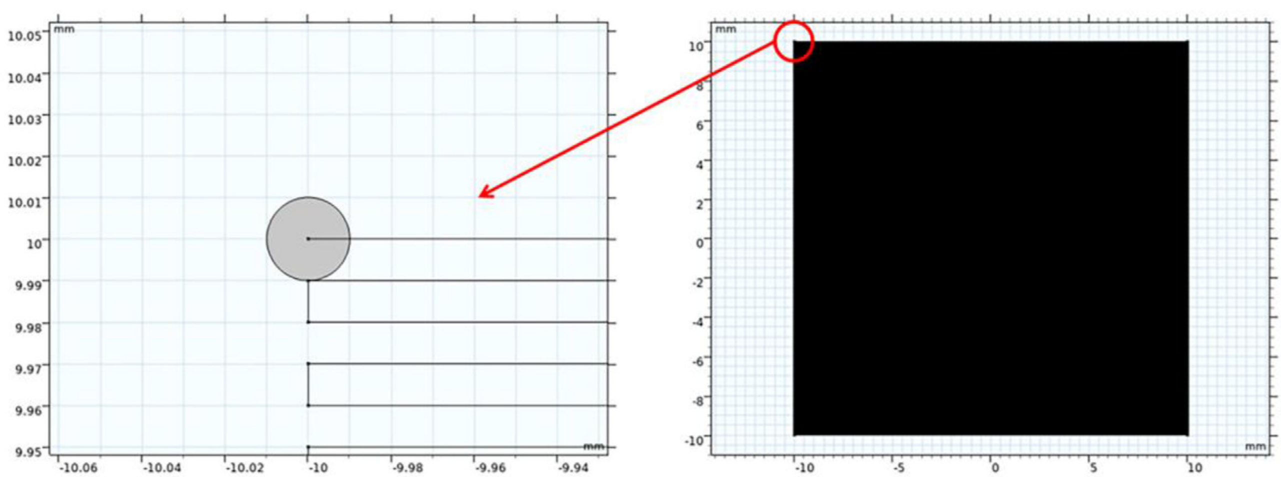

Fig. 2 – Femtosecond laser machining path diagram.

Conversely, if the number of effective pulses N is too high, the temperature cycles of the AgNWs may overlap, resulting in an excessive thermal effect. In order to strike a balance between femtosecond laser processing efficiency, the effective pulse number of the femtosecond laser, and the welding effectiveness of AgNWs, the scanning speed υ of the galvanometer was set at 100 mm/s in this experiment. By substituting this value into Eq. (1), the effective pulse number N was determined as 200. At this speed, the influence of femtosecond laser energy density on the morphology of AgNW electrodes was further studied.

3.1. *Effect of femtosecond laser energy density on morphology of AgNW electrodes*

Fig. 3 shows the SEM morphology of AgNW electrodes irradiated with different femtosecond laser energy densities. The femtosecond laser energy density F_0 were 0 $J\cdot m^{-2}$ (not irradiated), 1273 $J\cdot m^{-2}$, 2546 $J\cdot m^{-2}$, 5092 $J\cdot m^{-2}$, 7639 $J\cdot m^{-2}$ and 10,186 $J\cdot m^{-2}$, respectively. As shown in Fig. 3, there was no melting phenomenon in the AgNWs prepared by spinning coating at room temperature (Fig. 3(a)). With the increase of femtosecond laser energy density, the melting degree of AgNWs gradually increased (Fig. 3(b)~(F)), the fusion position of the AgNWs after welding was indicated by the dotted red box in the figure.

At a lower energy density of the femtosecond laser (1273 $J\cdot m^{-2}$), minimal changes were observed in most areas of the AgNW network. However, surface pre-melting was observed at the junction of two AgNWs, resulting in weak bonding between them (Fig. 3(b)). Surface pre-melting is a distinctive structural evolution behavior that occurs in metal nanomaterials. Due to the lack of coordinated atoms, the atoms on the surface of metal nanomaterials are held together by metal bonds, making them less stable compared to the atoms in the interior. Owing to the lack of fully coordinated atoms on the surface of metal nanomaterials, some metal bonds are left dangling, leading to reduced stability compared to the atoms in the interior. When the thermodynamic condition of solid-gas surface energy being greater than the sum of solid-liquid interfacial energy and liquid-gas interfacial energy is satisfied, the surface atoms of metal nanomaterials undergo disordering and partial melting, resulting in surface pre-melting before complete melting of the metal nanomaterials [25]. Surface pre-melting can occur even when the temperature is lower than the melting point of the metal nanomaterial, representing the initial stages of melting and facilitating subsequent shape transformations. It should be noted that surface pre-melting is not observed during the melting of macroscopic-scale (above micron) silver blocks and silver particles [26].

With an increase in femtosecond laser energy density (2546 $J\cdot m^{-2}$), the melting area on the AgNW network expanded, and the degree of melting also increased to some extent. The melting region was usually located at the overlapping region of two or more AgNWs, and the degree of melting rose with an increase in the number of AgNWs. Specifically, under this laser energy density, partial spheroidization of AgNWs was observed for the first time. As depicted in Fig. 3(c), the overlapping region consisted of four AgNWs.

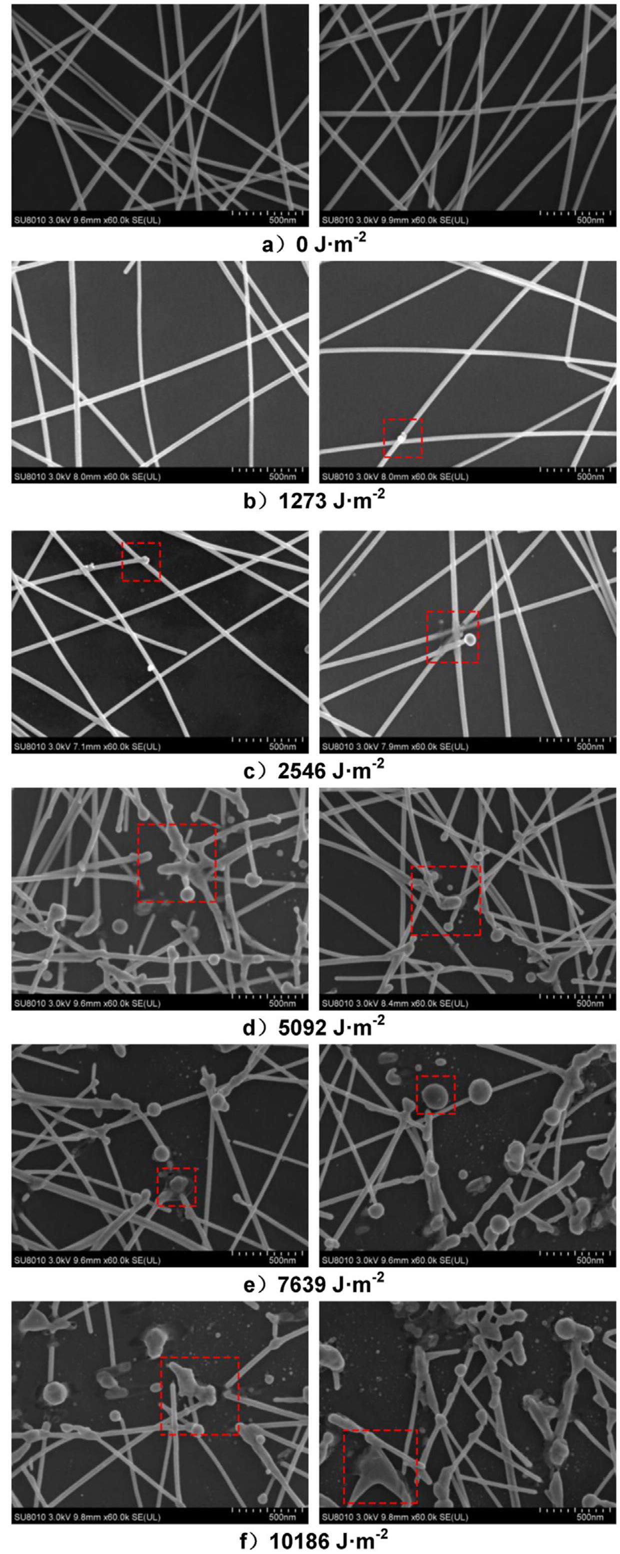

Fig. 3 – SEM morphologies of AgNW electrodes irradiated with different femtosecond laser energy densities.

When irradiated by a femtosecond laser, these four AgNWs were welded together, and the top end of the topmost AgNW underwent partial spheroidization. The resulting silver nanosphere had a diameter of approximately 90 nm, which was more than twice the diameter of the AgNWs. However, the silver nanosphere remained connected to the AgNW, preventing the disruption of the conductive pathway. Spheroidization, which seeks to minimize surface energy, is one of the most common failure modes in AgNWs. It involves the transformation of the nanowire from a cylindrical shape to a series of disconnected spheres under thermal stress. This process is driven by the decrease in melting point due to curvature (Gibbs-Thomson effect) and the morphological evolution mediated by surface diffusion (Rayleigh instability). The temperature at which AgNWs undergo spheroidization generally ranges from 250 to 350 °C [27–30]. This specific laser energy density (2546 $J \cdot m^{-2}$) could be regarded as the critical laser energy density for AgNW spheroidization. Although partial spheroidization occurred under the critical laser energy density, the resulting silver nanospheres remained connected to the AgNWs, leading to an enhanced welding effect between AgNWs.

With the further improvement of femtosecond laser energy density (5092 $J \cdot m^{-2}$), the melting scale of the AgNW network was further expanded. While femtosecond laser irradiation achieved partial welding of AgNWs, it also led to the formation of various-sized silver nanospheres on the AgNW network. The diameter of silver nanospheres formed by individual AgNW ranged from 40 to 50 nm, while at overlapping regions of multiple AgNWs, the diameter could reach up to 110 nm. Unfortunately, some silver nanospheres were unable to maintain connections with the surrounding AgNWs, resulting in their detachment from the AgNW network. This caused the failure of certain conductive pathways and led to a disruption in the AgNW network (Fig. 3(d)). Additionally, it was observed that the AgNWs melted mainly concentrated on the surface of the AgNW network, while the AgNWs inside the network had a relatively small degree of melting. In conclusion, under this laser energy density, although the welding of some AgNWs was achieved, some conductive pathways were also fused. The welding effect between AgNWs decreased to a certain extent.

With the further increase of femtosecond laser energy density (7639 $J \cdot m^{-2}$), the melting degree of the AgNW network rose again. Although femtosecond laser irradiation could achieve large-scale welding between AgNWs, the intense melting damaged the integrity of the AgNW network. While the number density of AgNWs decreased, some regions were replaced by large size silver nanospheres. These large-sized silver nanospheres with diameters close to 200 nm (Fig. 3(e)) were clearly not formed by the spheroidization of individual AgNWs but rather the result of multiple AgNWs undergoing excessive welding and fusion. Thus, the welding effect between AgNWs was not very ideal under this laser energy density.

With the further increase of femtosecond laser energy density (10,186 $J \cdot m^{-2}$), the substrate began to be damaged. After partial melting, local AgNWs might not even form silver nanospheres but instead transformed into irregularly shaped silver nanoblocks (Fig. 3(f)). These irregular silver nanoblocks had a maximum diameter of over 300 nm. At the same time, the effective length of AgNWs was greatly shortened, and some AgNWs were broken by the femtosecond laser into scattered silver nanochips on the substrate. Due to the large-scale failure of AgNWs, effective welding between AgNWs could not be achieved under this laser energy density.

In conclusion, femtosecond laser irradiation could realize welding between AgNWs, but the energy density of the femtosecond laser needed to be reasonably controlled. Considering that spheroidization failure of AgNWs can lead to the disruption of conductive pathways and damage the integrity of the AgNW network, the ideal welding effect is to achieve the soldering of the overlapping area of AgNWs before spheroidization failure occurs. Experimental results have shown that the critical laser energy density for AgNW spheroidization is approximately 2546 $J \cdot m^{-2}$. By using this critical energy density, not only effective welding between AgNWs can be achieved but also the integrity of the AgNW network can be preserved.

3.2. *Effect of femtosecond laser energy density on the sheet resistance of AgNW electrodes*

Sheet resistance is a critical metric for assessing the conductivity of AgNW electrodes, with its unit being $\Omega \cdot sq^{-1}$. A lower sheet resistance indicates enhanced charge transport performance along the plane direction of the electrode. Jia et al. defined the non-uniformity factor *NUF* (non-clothes factor) to evaluate the standard deviation of the resistance of thin electrode blocks [31]. The NUF calculation formula is shown as Eq. (2):

$$NUF = \sqrt{\frac{\sum_{i=1}^{n} (R_i - \overline{R})^2}{n\overline{R}^2}} \tag{2}$$

In the above equation, n represents the number of measurements, and R_i and R represent the measured value of the sheet resistance and the average value of all measurements, respectively. *NUF* can indirectly reflect the uniformity of the AgNW electrode. A smaller *NUF* value indicates a smaller standard deviation of the measured sheet resistance, indicating better uniformity of the AgNW electrode. Table 3 lists the measured values, average values and *NUF* of sheet resistance for AgNW electrodes at different femtosecond laser energy densities. Fig. 4 shows the variation of average sheet resistance and *NUF* with femtosecond laser energy density for AgNW electrodes. With the gradual increase of femtosecond laser energy density, the average sheet resistance of AgNW electrodes decreased first and then increases, while *NUF* increased first and then decreased and finally increased again.

When the femtosecond laser energy density F_0 was 1273 $J \cdot m^{-2}$, only local surface pre-melting occurred in the AgNW network. The welding effect between AgNWs was weak and the reduction of contact resistance was limited. The average sheet resistance decreased only slightly from the initial 39.28 $\omega \cdot sq^{-1}$ to 37.50 $\omega \cdot sq^{-1}$, the improvement of the conductivity of the AgNW electrodes was not significant. Simultaneously, due to less melting area, the uniformity of the AgNW electrodes decreased, and the *NUF* increased from 0.4% to 4.6%. When the femtosecond laser energy density F_0 was 2546 $J \cdot m^{-2}$, the melting degree of the AgNW network increased while the welding effect between AgNWs was enhanced. The average sheet resistance of the AgNW electrodes was further reduced to 18.89 $\omega \cdot sq^{-1}$. At the same time,

Table 3 – The measured values and average values of the sheet resistance of AgNW electrodes under different femtosecond laser energy densities.

Spinning coating process parameters	Initial sheet resistance $\Omega\cdot sq^{-1}$	Galvanometer scanning speed v mm/s	Laser energy density F_0 $J\cdot m^{-2}$	Measurement 1 of sheet resistance $\Omega\cdot sq^{-1}$	Measurement 2 of sheet resistance $\Omega\cdot sq^{-1}$	Measurement 3 of sheet resistance $\Omega\cdot sq^{-1}$	Average value of sheet resistance $\Omega\cdot sq^{-1}$	*NUF*%
2 mg/ml	39.28	100	1273	35.06	38.99	38.44	37.50	4.6
2000 rpm			2546	18.52	19.24	18.92	18.89	1.6
			5092	20.59	19.98	19.92	20.16	1.5
			7639	26.91	28.14	27.82	27.62	1.9
			10,186	51.59	40.51	41.13	44.41	11.4

the increase of melting region in the AgNW network improved the uniformity of the AgNW electrodes, so the *NUF* decreased to 1.6%. When the femtosecond laser energy density F_0 was 5092 $J\cdot m^{-2}$, the melting degree of the AgNW network was further increased. Although the welding of some AgNWs can reduce the contact resistance, the spheroidization of local AgNWs broke part of the conductive pathway. The negative effect caused by the melting of the conductive pathway was relatively large, so the average sheet resistance of the AgNW electrodes increased to 20.16 $\omega\cdot sq^{-1}$. Meanwhile, due to the relatively uniform melting of the AgNW network, there was not much variation in NUF, slightly decreasing to 1.5%. When the femtosecond laser energy density F_0 was 7639 $J\cdot m^{-2}$, the melting degree of the AgNW network was relatively drastic. More AgNWs were replaced by large size silver nanospheres, and the conductive pathway was seriously fused. The average sheet resistance of the AgNW electrodes rose further and increased to 27.62$\omega\cdot sq^{-1}$. In addition, due to the compromised integrity of the AgNW network, the uniformity decreased, resulting in an increase in NUF to 1.9%. When the femtosecond laser energy density F_0 was 10,186 $J\cdot m^{-2}$, the melting of the AgNW network was extremely severe, leading to widespread failure of the AgNWs and significant damage to the main conductive paths in the network. As a result, the average sheet resistance of the AgNW electrodes continued to increase, surpassing the initial value of 39.28 $\Omega\cdot sq^{-1}$, and reaching 44.41 $\Omega\cdot sq^{-1}$. Additionally, the damage to the substrate and the presence of irregularly shaped silver nanoclusters and silver nanoparticle debris severely disrupted the uniformity of the AgNW network. The *NUF* increased to 11.4%.

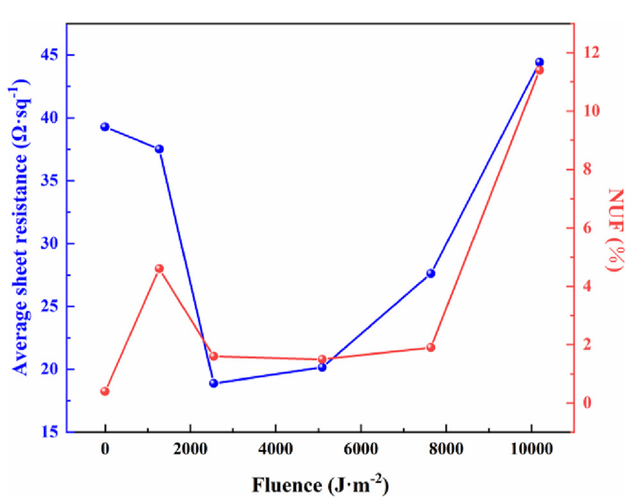

Fig. 4 – Average sheet resistance and *NUF* of AgNW electrodes with energy density.

In conclusion, as the femtosecond laser energy density increased, the conductivity of AgNW electrodes initially strengthened and then weakened, while uniformity exhibited a slight fluctuation followed by a decrease. The conductivity of the AgNW electrode is primarily constrained by the contact resistance between the AgNWs. In order to enhance the conductivity of the AgNW electrode, it is possible to sacrifice some of the conductive paths by performing a large-scale melting of the AgNW network while retaining the main conductive pathways. This approach can effectively reduce the contact resistance between the AgNWs and improve the overall conductivity of the AgNW electrode. When the femtosecond laser energy density F_0 was 2546 $J\cdot m^{-2}$, the average sheet resistance of AgNW electrodes was the smallest, only 18.89 $\omega\cdot sq^{-1}$, and the *NUF* was also small, 1.6%. Which means that after the femtosecond laser irradiation with this energy density, the conductivity of AgNWs had been greatly improved and the uniformity had been maintained.

3.3. *Effect of femtosecond laser energy density on transmittance of AgNW electrodes*

Fig. 5 shows the transmittance curve of the AgNW electrodes as a function of femtosecond laser energy density, and Fig. 6 displays the transmittance curve of AgNW electrodes at 550 nm with femtosecond laser energy density, which corresponds to the wavelength of maximum human visual sensitivity [32–37]. The femtosecond laser energy densities F_0 were 0 $J\cdot m^{-2}$ (not irradiated), 1273 $J\cdot m^{-2}$, 2546 $J\cdot m^{-2}$, 5092 $J\cdot m^{-2}$, 7639 $J\cdot m^{-2}$ and 10,186 $J\cdot m^{-2}$, respectively.

According to Fig. 5, with the increase of femtosecond laser energy density, the transmittance curve of AgNW electrodes decreased first and then increased. Compared to the initial transmittance curve, the changes in transmittance at various wavelengths after femtosecond laser irradiation were relatively small, with variations generally staying within 2%.

From Fig. 6, The variation of the transmittance of AgNW electrodes at 550 nm after irradiation with femtosecond laser at different energy densities was generally within 1%. Before the femtosecond laser energy density (F_0) reached the critical value of 2546 $J\cdot m^{-2}$, the transmittance at 550 nm of the AgNW electrodes gradually and slowly decreased. However, once the F_0 exceeded the critical value and caused the AgNWs to

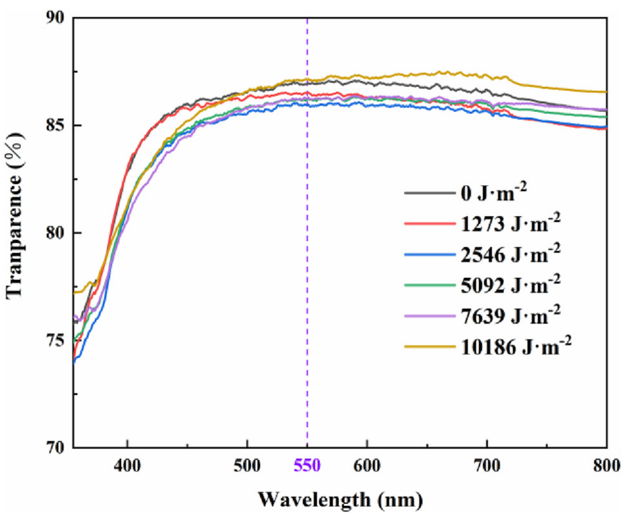

Fig. 5 – The curve of the variation in transmittance with femtosecond laser energy density.

undergo spherical transformation, the transmittance at 550 nm started to slowly rise again. The former was due to the fact that at lower energy densities, the locally melted AgNWs spread outwards, increasing the surface area of the silver on the transparent substrate, which hindered the transmission of light. Additionally, the filling of gaps between AgNWs closed off some of the originally penetrable light paths. As a result, the transmittance of the AgNW electrodes slightly decreased. The latter was attributed to the phenomenon of spherical transformation occurring in localized melted AgNWs under high energy density, driven by surface tension. As the volume remains constant, a sphere has the smallest surface area. The spherical transformation of AgNWs alleviated the spreading of liquid silver on the transparent substrate and reduced the surface area of silver. Consequently, the transmittance of the AgNW electrodes exhibited a slight recovery.

In conclusion, with the gradual increase of femtosecond laser energy density, although the transmittance of AgNW electrodes decreased first and then increased, the range of change was small. Which meant that femtosecond laser irradiation had little effect on the transmittance of AgNW electrodes. When the femtosecond laser energy density F_0 was 2546 $J{\cdot}m^{-2}$, the transmittance of the AgNW electrodes at 550 nm was relatively the smallest, which was 85.96%.

3.4. *Effect of femtosecond laser energy density on quality factors of AgNW electrodes*

Due to the inherent contradiction between conductivity and transmittance of the AgNW transparent electrodes, the quality factor *FoM* is introduced to reflect the photoelectric characteristics of AgNW electrodes. *FoM* is defined as the ratio of the DC conductivity of the AgNW electrodes and the photo-induced conductivity, namely σ_{DC}/σ_{OP}. The higher the value of *FoM*, the better the photoelectric performance of the electrode [38]. For AgNW electrodes to be used as commercial electrodes, the *FoM* must be greater than 35. The relationship between the transmittance T, the sheet resistance R_s and the quality factor *FoM* of the AgNW electrodes can be expressed by the following equation [39]:

$$T=\left[1+\frac{Z_0}{2R_S}\frac{\sigma_{OP}}{\sigma_{DC}}\right]^{-2} \tag{3}$$

where T is the transmittance of the electrode at 550 nm, Z_0 is the free-space impedance, whose value is 377Ω, and R_s is the sheet resistance of the electrode. The above equation can be simplified as follows:

$$FoM=\frac{188.5\sqrt{T}}{(1-\sqrt{T})R_S} \tag{4}$$

The *FoM* for each electrode can be calculated by substituting the sheet resistance (R_s) and transmittance (T) at 550 nm into Eq. (4). Fig. 7 shows the variation of the quality factor *FoM* of AgNW electrodes with the femtosecond laser

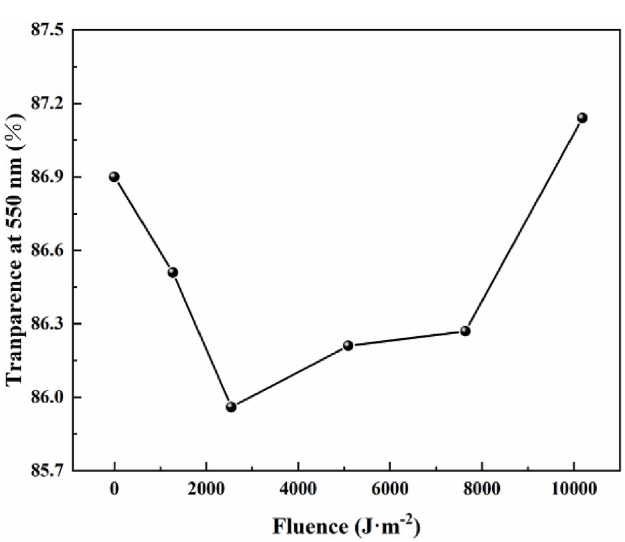

Fig. 6 – The curve of the variation in transmittance at 550 nm with femtosecond laser energy density.

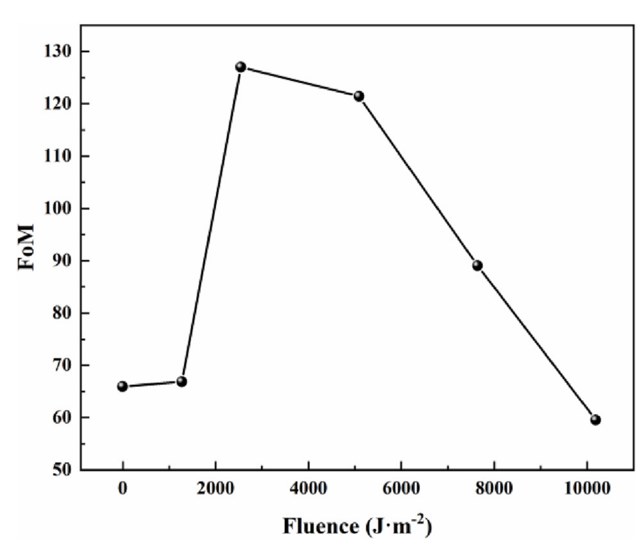

Fig. 7 – The curve of the variation in *FoM* with femtosecond laser energy density.

energy density. The femtosecond laser energy densities F_0 were 0 $J\cdot m^{-2}$ (not irradiated), 1273 $J\cdot m^{-2}$, 2546 $J\cdot m^{-2}$, 5092 $J\cdot m^{-2}$, 7639 $J\cdot m^{-2}$ and 10,186 $J\cdot m^{-2}$, respectively. It could be found from Fig. 5 that the value of *FoM* first increased and then decreased. When the femtosecond laser energy density F_0 was 2546 $J\cdot m^{-2}$, the value of the quality factor *FoM* reached a maximum of 126.99, nearly twice the initial value, which fully meet the needs of commercial electrodes.

In conclusion, it could be seen that the quality factor *FoM* of AgNW electrodes was mainly affected by the sheet resistance. The quality of AgNW electrodes could be effectively improved after femtosecond laser irradiation with appropriate energy density. Indeed, it is important to note that excessively high energy density can lead to a rapid decline in the quality of the AgNW electrodes.

3.5. *Effect of femtosecond laser effective pulse number on welding effect of AgNWs*

In the previous discussion, it was assumed that a larger effective pulse number *N* may result in the superposition of temperature cycles in the AgNWs, leading to significant thermal effects. On the other hand, a smaller effective pulse number *N* may not achieve sufficient welding between the AgNWs. Therefore, an effective pulse number of 200 was set, and the femtosecond laser energy density was optimized based on this value. In order to validate the hypothesis, the influence of the number of effective femtosecond laser pulses on the welding effect of AgNWs was further studied by adjusting the scanning speed υ based on the optimized femtosecond laser energy density ($F_0 = 2546\ J\cdot m^{-2}$). This study aimed to explore in more depth the relationship between the efficiency of femtosecond laser processing and the quality factors of AgNW electrodes.

To quantify the femtosecond laser processing efficiency, the time required for complete irradiation of a 20 mm × 20 mm sample at different galvanometer scanning speeds was recorded during the experiment. The results showed that when υ was 1000 mm/s ($N = 20$), the time required for complete irradiation was 48 s. At a scanning speed of $\upsilon = 100$ mm/s ($N = 200$), the time required was over 7 min. When υ was 10 mm/s ($N = 2000$), it took over 1 h to complete irradiation.

Fig. 8 illustrates the variation of the average sheet resistance and *NUF* of the AgNW electrodes with the number of effective pulses. Fig. 9 displays the variations of transmittance and the quality factor, *FoM*, of the AgNW electrodes at 550 nm with the number of effective pulses. The experimental results revealed that as the number of effective pulses increases from 20 to 200 and then to 2000, the average sheet resistance of the AgNW electrodes decreased from 21.71 $\Omega\cdot sq^{-1}$ to 18.89 $\Omega\cdot sq^{-1}$ and then increased to 23.01 $\Omega\cdot sq^{-1}$. The *NUF* decreased from 1.9% to 1.6% and then to 1.2%. Transmittance initially decreased from 86.89% to 85.96% and then increased to 86.55%. The quality factor, *FoM*, firstly increased from 119.12 to 126.99 and then decreased to 109.38.

It could be found that compared with adjusting femtosecond laser energy density, changing the effective pulse number *N* by adjusting galvanometer scanning speed υ did not have a great impact on the welding effect of AgNWs. Even though the effective pulse number *N* increased by orders of magnitude, the melting degree of AgNWs and the indicators of AgNW electrodes did not fluctuate significantly.

In conclusion, the hypothesis that a larger number of effective pulses, *N*, can lead to the superposition of temperature cycles in AgNWs, resulting in a greater thermal effect, while a smaller number of effective pulses, *N*, may not

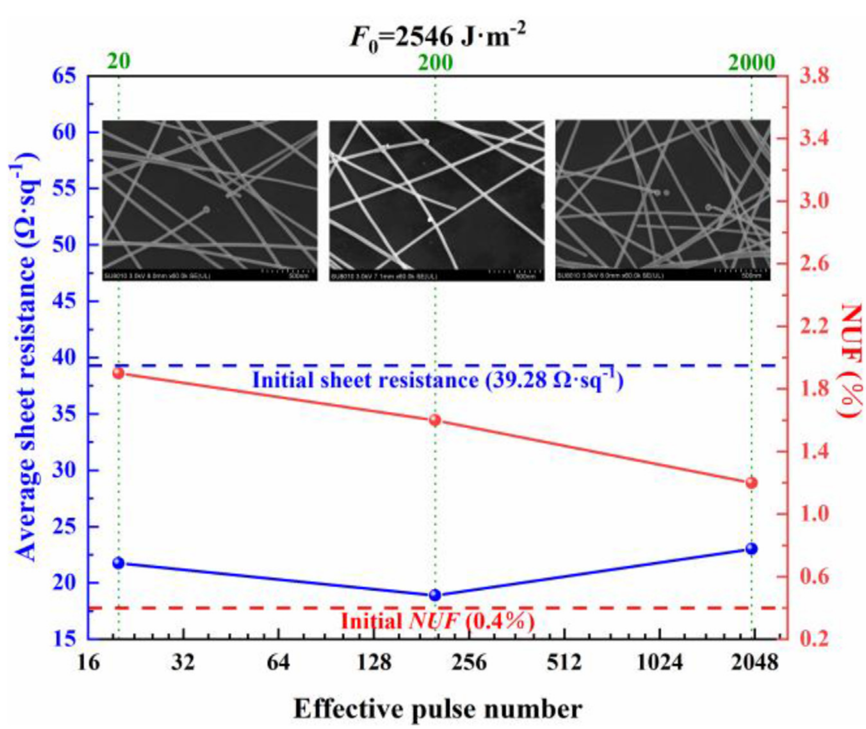

Fig. 8 – The curve of the variation in sheet resistance and *NUF* with effective pulse number.

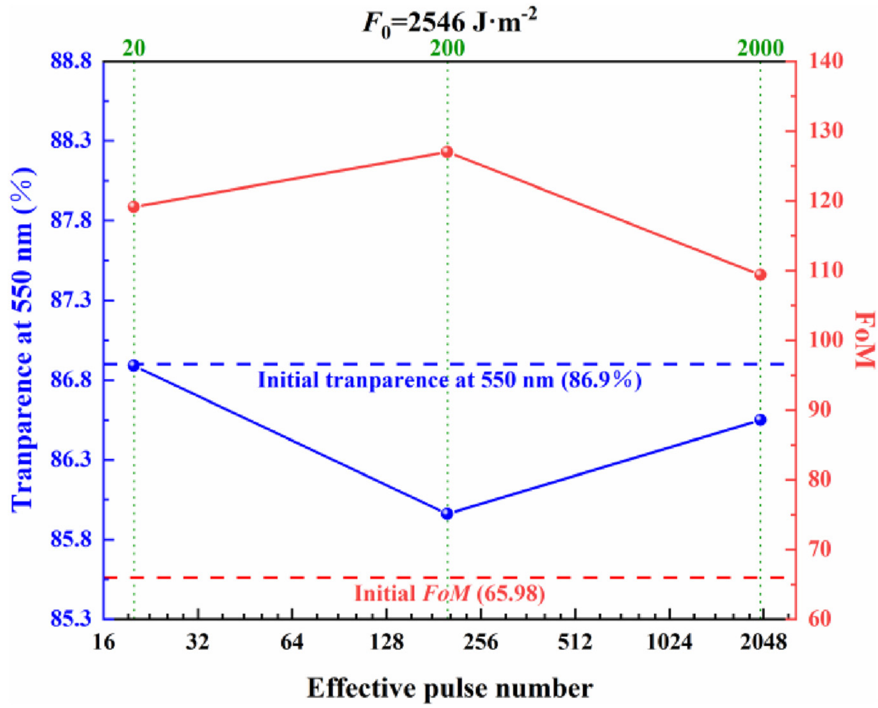

Fig. 9 – The curve of the variation in transmittance and *FoM* at 550 nm with effective pulse number.

achieve sufficient welding between AgNWs, is not valid. It seems that the temperature cycles in AgNWs under the action of each femtosecond laser pulse are independent events and do not undergo superposition. When the galvanometer scanning speed v was 100 mm/s, the femtosecond laser welding of AgNWs had the best effect and higher processing efficiency. The effective pulse number of femtosecond laser had little influence on the welding effect of AgNWs. If there are many samples to be processed, a small part of quality factors can be slightly sacrificed to pursue higher processing efficiency by increasing the galvanometer scanning speed to 1000 mm/s.

4. Conclusions

(1) Femtosecond laser could achieve welding between AgNWs, but it was necessary to control the energy density of the laser reasonably. Too small energy density would have limited impact on the AgNW network, while excessively high energy density could lead to extensive spheroidization of the AgNWs and breakage of some conductive paths. With the increase of femtosecond laser energy density, the conductivity of the AgNW electrodes initially enhanced and then weakened. The sheet resistance could decrease by up to 52%, while the uniformity of the AgNW electrodes decreased after a slight fluctuation. The transmittance of the AgNW electrodes was not significantly affected by femtosecond laser irradiation, with a change in transmittance at 550 nm within 1%. Overall, the *FoM*, which represents the quality of the AgNW electrodes, initially increased and then decreased as the energy density of the femtosecond laser increased.

(2) When the energy density of the femtosecond laser, F_0, reached the critical value of 2546 J·m^{-2} for spheroidization of the AgNWs, effective welding between the nanowires was achieved while avoiding extensive spheroidization and breakage of the conductive paths. At this point, the average sheet resistance of the AgNW electrodes was only 18.89 Ω·sq^{-1}, with a non-uniformity factor (*NUF*) of 1.6% and a transmittance of 85.96% at 550 nm. The *FoM* was close to twice the initial value, reaching 126.99.

(3) The number of effective pulses of the femtosecond laser did not have a significant impact on the welding effect of the AgNWs. Even with a substantial increase in the number of effective pulses, the degree of melting of the AgNWs and the various indicators of the AgNW electrodes did not show significant fluctuations. The best welding effect of the femtosecond laser on the AgNWs was achieved when the scanning speed of the galvanometer was 100 mm/s. This resulted in the most apparent improvement in the comprehensive performance of the AgNW electrodes, while also ensuring high processing efficiency (total irradiation time of about 7 min).

CRediT authorship contribution statement

Cheng Chen: Conceptualization, Investigation, Visualization, Writing - Original Draft. **Xiaopeng Li:** Methodology, Supervision, Funding acquisition, Writing - Review & Editing.

Jiayue Zhou: Investigation, Formal analysis. **Hongxing Li:** Funding acquisition. **Hongxia Lin:** Funding acquisition. **Yong Peng:** Project administration. **Kehong Wang:** Resources, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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